

TrES-3b

R-2803

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_150723 · created 2026-09-05T00:18:40+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457226.7119 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.056
Transit depth (Rp/Rs)^2	3.84 +/- 2.2 [%]
Orbital Inclination (inc)	86.84 +/- 2.91
Airmass coefficient 1 (a1)	1.02 +/- 0.21
Airmass coefficient 2 (a2)	-0.07 +/- 1.54
Scatter in the residuals of the lightcurve fit is	21.36 %
Best Comparison Star	#1 - [291, 249]
Optimal Aperture	51.58
Optimal Annulus	128.96
Transit Duration (day)	0.08 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.196 ± 0.056 vs 0.16309 (deviation 0.33σ)
Duration fit vs published	0.08 d vs 0.0567 d (deviation 1.19σ)
Mid-time O–C	-2.0 min
Depth SNR	2.0
Residual scatter	21.36 %

Light curve

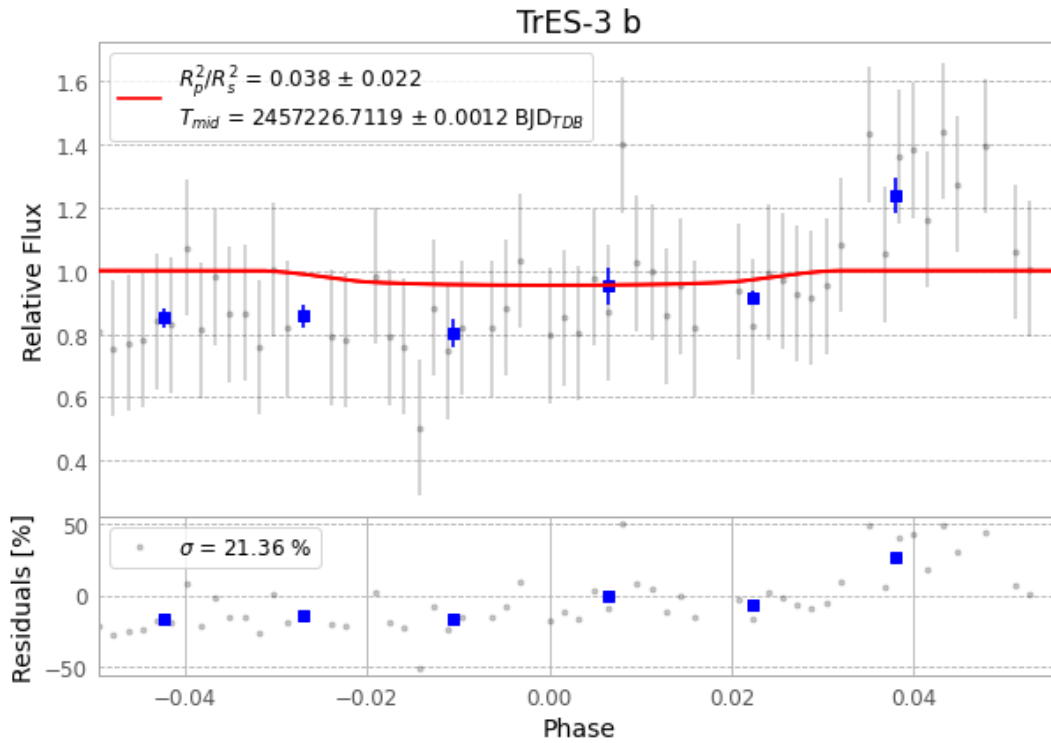


Figure 1 — FinalLightCurve_TrES-3 b_2015-07-22.png (SHA-256 c9c630f0ba12ea6b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [325, 254], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8807ab1cfdd97eeb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

67 FITS frames (44 MB), TRES-3150723032712.FITS → TRES-3150723064512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-150723.log (1e51eab755b9...), FinalLightCurve_TrES-3 b_2015-07-22.png (c9c630f0ba12...), FinalLightCurve_TrES-3 b_2015-07-22.csv (bd7a5fff3a70...)

Compute resources

Wall time 954.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 00:18Z.